🌱 PLANT NUTRITION — FULL MIND MAP

Biology Ch. 8.ii · Plant Physiology · Hinglish · SSC / BPSC / BSSC / Railway

Topics in this sub-chapter:

1. Plant Parts & Stem Functions — Shrub, Tree, Creeper & Petiole

- Petiole leaf ko stem se attach karta hai aur weak stem jo upright nahi reh sakta, creeper hota hai. Source ke matching question mein base ke paas branching wala thick, hard stem "Tree" nahi balki Shrub bataya gaya hai; tree mein branches trunk par comparatively upar hoti hain. *[Q1]*
- Stem plant ko shape/support deta hai, leaves se food ko plant ke doosre parts tak transport karta hai aur food store bhi kar sakta hai. Green stems photosynthesis bhi kar sakte hain; isi wajah se source question mein "more than one" framing apply hoti hai. *[Q2]*

⚠ EXAM TRAP — Shrub vs Tree

- ▸ Thick, hard stem + branching near base = SHRUB; Tree mein branches usually ground level se upar hoti hain. *[Q1]*
- ▸ Petiole = leaf ko stem se jodta hai; Creeper = weak stem, upright stand nahi kar pata. *[Q1]*

2. Root Hairs, Stomata & Transpiration Pull

- Plant ko required maximum water root hairs absorb karte hain. Ye root ki outer layer/cell-maturation zone ke tiny hair-like extensions hote hain jo absorption ke liye surface area badhate hain. *[Q3]*
- Stomata leaf epidermis ke small pores hote hain; guard cells stomatal opening ko regulate karke atmosphere ke saath gas exchange mein help karte hain. Transpiration bhi mainly stomata ke through hota hai. *[Q4]*
- Stomata ka open/close hona guard cells mein water amount aur usse related turgor pressure ke change par depend karta hai. Water gain se guard cells turgid hokar pore open karte hain; water loss se shrink hokar pore close hota hai. *[Q5]*
- Tall trees mein water great heights tak transpiration se create hone wale suction/negative pressure yani transpiration pull ke karan xylem ke through upar reach karta hai. *[Q6]*

⚠ EXAM TRAP — Stomata vs Transpiration Pull

- ▸ Stomata opening/closing = Guard-cell water/turgor change. *[Q5]*
- ▸ Water ko tree mein upar kheenchne wali force = Transpiration pull. *[Q6]*

3. Xylem vs Phloem — Water, Minerals & Food Transport

- Tree ki bark ko base ke paas circularly remove karne (girdling) se phloem damage/remove ho jata hai. Leaves ka photosynthetic food roots tak nahi pahunchta, roots starve karte hain aur plant gradually dry/die kar sakta hai. *[Q7]*
- Xylem vascular plants mein roots se water aur soluble mineral nutrients ko stem/leaves aur plant ke doosre parts tak transport karta hai. Source ke repeated questions mein water transport ka answer Xylem hi hai. *[Q8, Q9, Q10]*

- Phloem (bast tissue) leaves mein bane food/photosynthates aur organic nutrients ko plant ke doosre parts tak carry karta hai. Source xylem ko primarily dead cells aur phloem ko living cells se bana hua batata hai. *[Q11, Q12]*

⚠ EXAM TRAP — Xylem vs Phloem

- ▸ XYLEM = Water + minerals, mainly roots → upper plant parts. *[Q8–Q10]*
- ▸ PHLOEM = Prepared food / organic nutrients; living transport tissue. *[Q11, Q12]*
- ▸ Girdling bark removes/damages phloem → roots ko food supply rukti hai. *[Q7]*

4. Plant Growth — Why a Nail Stays at the Same Height

- Tree trunk mein soil level se 1 m height par lagaya nail years baad bhi approximately same height par rahega. Length growth shoot/root tips ke apical meristems se hoti hai, jabki trunk ka girth lateral meristems se badhta hai; nail upward move nahi karta. *[Q13]*

5. Essential Plant Nutrients — Soil, Macro & Micro Elements

- Source ke according plants apne nutrients mainly soil se receive karte hain. Carbon, oxygen aur hydrogen air/water se milte hain, lekin nitrogen, phosphorus, potassium jaise bahut se mineral nutrients roots soil se absorb karte hain. *[Q14]*
- Sodium ko source most plants ke growth ke liye generally essential element nahi maanta. Potassium, calcium aur magnesium essential macronutrients hain; sodium kuch specific plants mein beneficial ho sakta hai. *[Q15]*
- Potassium (K) macronutrient hai, jabki Zinc, Boron aur Chlorine micronutrients/trace elements hain. *[Q16]*
- Magnesium micronutrient nahi, macronutrient hai aur chlorophyll molecule ka central component bataya gaya hai. Iron, Manganese aur Copper micronutrients hain. Sodium ko source essential micronutrient bhi nahi maanta; Boron, Zinc aur Copper essential micronutrients hain. *[Q17, Q18]*

⚠ EXAM TRAP — Macro vs Micro Nutrients

- ▸ Potassium + Magnesium = MACRONUTRIENTS. *[Q16, Q17]*
- ▸ Boron, Zinc, Copper, Iron, Manganese, Chlorine = source ke micronutrient examples. *[Q16–Q18]*
- ▸ Sodium = most plants ke liye generally essential element/micronutrient NAHI. *[Q15, Q18]*

6. Sulphur, Nitrogen, Phosphorus & Boron — Crop Nutrition

- Mustard crop mein oil content enhance karne ke liye Sulphur ko source most useful nutrient batata hai; sulphur plant metabolism aur seed oil synthesis mein important hai. *[Q19]*
- Plants atmospheric N_2 ko directly normally use nahi karte; source ke repeated questions mein nitrogen uptake ka primary/common form Nitrate (NO_3^-) diya gaya hai. *[Q20, Q21, Q22]*
- Early plant growth stimulate karne ke liye phosphorus-containing fertilizer useful hai. Source phosphorus ko root development, early establishment aur maturity hasten karne se link karta hai. *[Q23]*
- Legumes mein nodule formation ke liye Boron necessary bataya gaya hai. Boron new-cell/meristem development ke saath sugars, starches, nitrogen aur phosphorus ke translocation se bhi associated hai. *[Q24]*

⚠ EXAM TRAP — Nitrate, Phosphorus & Boron

- ▸ Plant nitrogen uptake (source framing) = NITRATE, NO_3^-. *[Q20–Q22]*
- ▸ Early root/plant growth = PHOSPHORUS. *[Q23]*
- ▸ Legume nodule formation = BORON. *[Q24]*

7. Parasitic Nutrition — Cuscuta (Amarbel)

- Cuscuta/Amarbel ek yellow-orange/red, thin-stemmed parasitic plant hai; leaves minute scales tak reduced hoti hain. Ye haustoria bana kar host se nutrients leta hai. Source Pitcher plant aur Bladderwort ko carnivorous, parasite nahi batata hai. *[Q25]*
- Assertion–Reason question mein "Cuscuta parasitic angiosperm hai" true hai, lekin "nutrition host leaves se leta hai" false mark hai. Source ke mutabik haustoria host stem ke xylem aur phloem se contact karke prepared food, water aur minerals absorb karte hain. *[Q26]*

⚠ EXAM TRAP — Cuscuta Nutrition

- ▸ Cuscuta = Parasitic angiosperm. *[Q25, Q26]*
- ▸ Nutrition sirf host leaves se nahi; haustoria host STEM ke xylem + phloem se connect karte hain. *[Q26]*

8. Crop Logging — Nutrient Requirement Assessment

- Crop-logging plant analysis ki method hai jisse crop production ke liye nutrient requirement assess ki jaati hai, taaki good plant growth aur high yield achieve ho. Source concept ko H.F. Clements (USA) aur Hawaii mein sugarcane cultivation se associate karta hai. *[Q27]*